

TrES-5b

R-1832

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_210823 · created 2026-08-03T05:57:45+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2459449.7602 +/- 0.0032 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.185 +/- 0.042
Transit depth (Rp/Rs)^2	3.42 +/- 1.55 [%]
Orbital Inclination (inc)	83.34 +/- 2.29
Airmass coefficient 1 (a1)	1.0 +/- 0.036
Airmass coefficient 2 (a2)	0.01 +/- 0.13
Scatter in the residuals of the lightcurve fit is	3.6 %
Best Comparison Star	#3 - [128, 71]
Optimal Method	PSF photometry
Transit Duration (day)	0.068 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.185 ± 0.042 vs 0.142 (deviation 1.02σ)
Duration fit vs published	0.068 d vs 0.0758 d (deviation 0.39σ)
Mid-time O–C	-8.1 min
Depth SNR	4.4
Residual scatter	3.6 %

Light curve

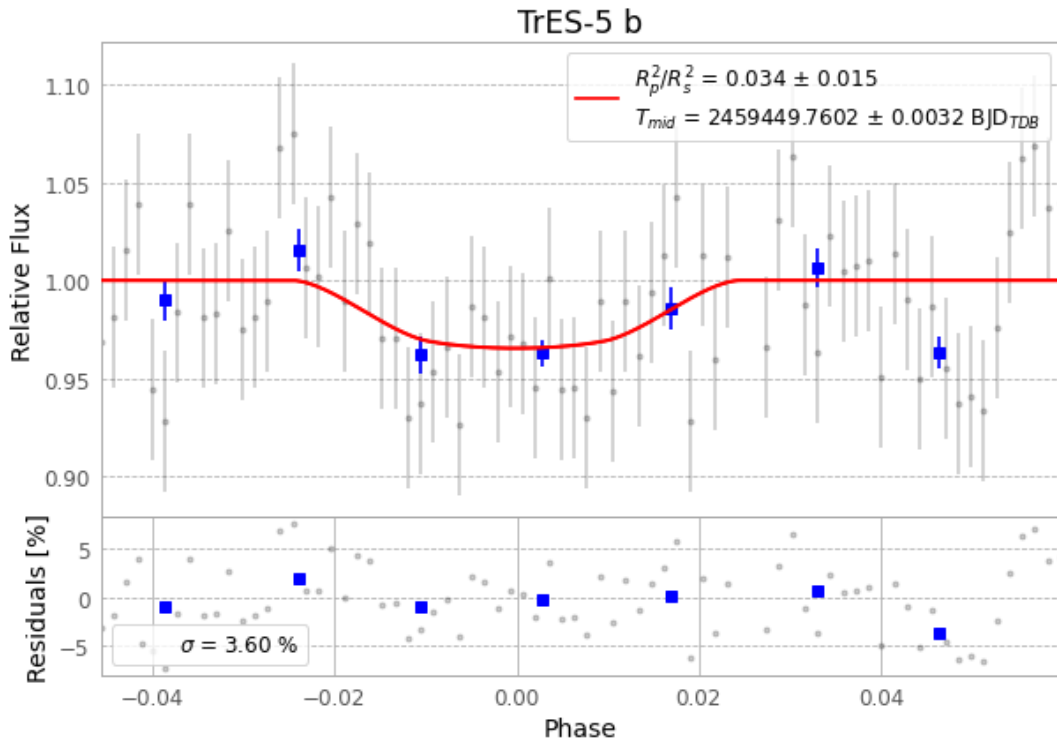


Figure 1 — FinalLightCurve_TrES-5 b_2021-08-22.png (SHA-256 9627ca660ae4de0e...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [421, 223], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bebb24100304d78e...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

74 FITS frames (49 MB), TRES-5210823043312.FITS → TRES-5210823081812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-210823.log (7d63d1bc1367...), FinalLightCurve_TrES-5 b_2021-08-22.png (9627ca660ae4...), FinalLightCurve_TrES-5 b_2021-08-22.csv (676a8252f4c4...)

Compute resources

Wall time 180.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-03 05:57Z.